

t

- 14 B** At constant P , $V \propto T$. When cooled, T decreases, so V decreases. Since gas undergoes compression, work is done on the gas. $W = +ve$

When T decreases, $\Delta U = -ve$,

according to 1st law of thermodynamics $\Delta U = q + W$

$$-ve = q + (+ve) \Rightarrow q \text{ must be } -ve$$

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D

Given $x_0 = 0.030 \text{ m}$ from graph $\rightarrow$ period $T = 2.0 \text{ s}$